$$
\begin{aligned}
C_{\phi W}^{(22)} \to \frac{1}{16\,\pi^2}\, \Bigg(& -\frac{1}{16}\, g_2{}^2 \left(2\, \overline{y_e}^{pr}\, y_e{}^{pr} + \textstyle\sum_p g_1{}^2\right) LF_{3,0}\left[m_{\tilde{l}}{}^p\right] + \frac{1}{16}\, g_2{}^2 \left(2\, \overline{y_e}^{pr}\, y_e{}^{pr} + \textstyle\sum_p g_1{}^2\right) LF_{4,-1}\left[m_{\tilde{l}}{}^p\right] + \\
& \frac{1}{16}\, g_2{}^2 \left(-6\, \overline{y_d}^{pr}\, y_d{}^{pr} + \textstyle\sum_p g_1{}^2\right) LF_{3,0}\left[m_{\tilde{q}}{}^p\right] + \frac{1}{16}\, g_2{}^2 \left(6\, \overline{y_d}^{pr}\, y_d{}^{pr} - \textstyle\sum_p g_1{}^2\right) LF_{4,-1}\left[m_{\tilde{q}}{}^p\right] + \\
& \frac{5}{4}\, g_2{}^4\, LF_{2,2,-1}[m_2,\, \tilde{\mu}] + g_2{}^4\, LF_{3,1,-1}[m_2,\, \tilde{\mu}] - g_2{}^4\, LF_{3,2,-2}[m_2,\, \tilde{\mu}] - g_2{}^4\, LF_{4,1,-2}[m_2,\, \tilde{\mu}] - \\
& \frac{1}{4}\, g_2{}^2\, \overline{a_e}^{pr}\, a_e{}^{pr}\, LF_{4,1,-1}\left[m_{\tilde{l}}{}^p,\, m_{\tilde{e}}{}^r\right] + \frac{1}{4}\, g_2{}^2\, \overline{a_e}^{pr}\, a_e{}^{pr}\, LF_{5,1,-2}\left[m_{\tilde{l}}{}^p,\, m_{\tilde{e}}{}^r\right] - \\
& \frac{3}{4}\, g_2{}^2\, \overline{a_d}^{pr}\, a_d{}^{pr}\, LF_{4,1,-1}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{d}}{}^r\right] + \frac{3}{4}\, g_2{}^2\, \overline{a_d}^{pr}\, a_d{}^{pr}\, LF_{5,1,-2}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{d}}{}^r\right] - \\
& \frac{3}{8}\, g_2{}^2\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{2,2,0}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{u}}{}^r\right] - \frac{3}{8}\, g_2{}^2\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{3,1,0}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{u}}{}^r\right] + \\
& \frac{3}{8}\, g_2{}^2\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{3,2,-1}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{u}}{}^r\right] + \frac{3}{8}\, g_2{}^2\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{4,1,-1}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{u}}{}^r\right] + \\
& \frac{3}{8}\, g_2{}^2\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{3,1,0}\left[m_{\tilde{u}}{}^r,\, m_{\tilde{q}}{}^p\right] + \frac{3}{8}\, g_2{}^2\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{3,2,-1}\left[m_{\tilde{u}}{}^r,\, m_{\tilde{q}}{}^p\right] - \\
& \frac{9}{8}\, g_2{}^2\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{4,1,-1}\left[m_{\tilde{u}}{}^r,\, m_{\tilde{q}}{}^p\right] + \frac{3}{4}\, g_2{}^2\, \tilde{\mu}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{5,1,-2}\left[m_{\tilde{u}}{}^r,\, m_{\tilde{q}}{}^p\right] - \\
& \frac{1}{8}\, g_1{}^2\, g_2{}^2\, LF_{3,1,-1}[\tilde{\mu},\, m_1] + \frac{3}{8}\, g_1{}^2\, g_2{}^2\, LF_{4,1,-2}[\tilde{\mu},\, m_1] - \frac{1}{4}\, g_1{}^2\, g_2{}^2\, LF_{5,1,-3}[\tilde{\mu},\, m_1] - \\
& \frac{3}{8}\, g_2{}^4\, LF_{3,1,-1}[\tilde{\mu},\, m_2] - g_2{}^4\, LF_{3,2,-2}[\tilde{\mu},\, m_2] + \frac{9}{8}\, g_2{}^4\, LF_{4,1,-2}[\tilde{\mu},\, m_2] - \frac{3}{4}\, g_2{}^4\, LF_{5,1,-3}[\tilde{\mu},\, m_2] \Bigg)
\end{aligned}
$$